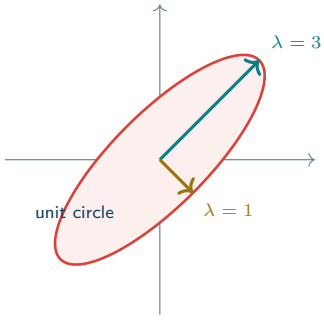
LINEAR ALGEBRA LADDER 7 OF 9

Eigenvalues and eigenvectors

An eigenvector keeps its direction under a matrix and is only stretched, by a factor called the eigenvalue.

1 / THE RULE AND THE PICTURE

A matrix turns the unit circle into an ellipse whose axes are its eigenvectors



For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ the circle becomes an ellipse with semi-axes 3 along $(1,1)$ and 1 along $(1,-1)$, because

$$Av = \lambda v: \quad A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Along these two directions the matrix acts as plain multiplication by 3 and by 1.

2 / FINDING THEM / THE CHARACTERISTIC EQUATION

The eigenvalues are the roots of $\det(A - \lambda I) = 0$, a quadratic in λ

Here $\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = (1-\lambda)(3-\lambda)$, a difference of squares, so $\lambda = 1$ and $\lambda = 3$. Expanding, $\lambda^2 - 4\lambda + 3 = 0$, whose coefficients are the trace $2 + 2 = 4$ and the determinant $4 - 1 = 3$: the eigenvalues add to the trace and multiply to the determinant.
Each eigenvector solves $(A - \lambda I)v = 0$, for example $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0$ gives $v = (1,1)$ at $\lambda = 3$.

3 / A GALLERY / WHAT EIGENVALUES LOOK LIKE

Some matrices have two real eigen-directions, some one, and some none

matrix	trace	det	eigenvalues	eigenvectors
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	4	3	$3,\ 1$	$(1,1),\ (1,-1)$
stretch $\mathrm{diag}(2,1)$	3	2	$2,\ 1$	axes
projection $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	1	0	$1,\ 0$	x-axis, y-axis
reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	0	-1	$1,\ -1$	axes
shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	2	1	$1,\ 1$	only $(1,0)$
rotation $90°$	0	1	$\pm i$	none real

The shear has the repeated eigenvalue 1 but only one direction, so it cannot be diagonalised, and the rotation turns every direction, so its eigenvalues $\pm i$ are complex.

4 / POWERS / WHY THEY MATTER

Repeated application multiplies each eigen-direction by λ^n

Since $A\,v = \lambda v$ we have $A^n v = \lambda^n v$, so $A^2(1,1) = 9(1,1)$. Any vector splits along the eigenvectors, $(3,1) = 2(1,1) + 1(1,-1)$, and then $A^n(3,1) = 2 \cdot 3^n(1,1) + 1 \cdot 1^n(1,-1)$; for $n = 1$ this is $(6+1,\ 6-1) = (7,5) = A(3,1)$.
After many steps the direction with the largest $|\lambda|$ dominates, which is why the iteration settles along $(1,1)$.

ASIDE / THE QUADRATIC AGAIN

The two eigenvalues are where the line $u + v = \mathrm{trace}$ meets the hyperbola $uv = \det$, the picture for any quadratic.

LINKS TO OTHER SHEETS

The characteristic equation is developed in *The characteristic equation beneath five presentations*; the previous sheet is *Span, basis and the four subspaces* and the next is *Diagonalisation and powers*.

Convention. λ is an eigenvalue and $v \neq 0$ an eigenvector; trace is the sum of the diagonal entries.